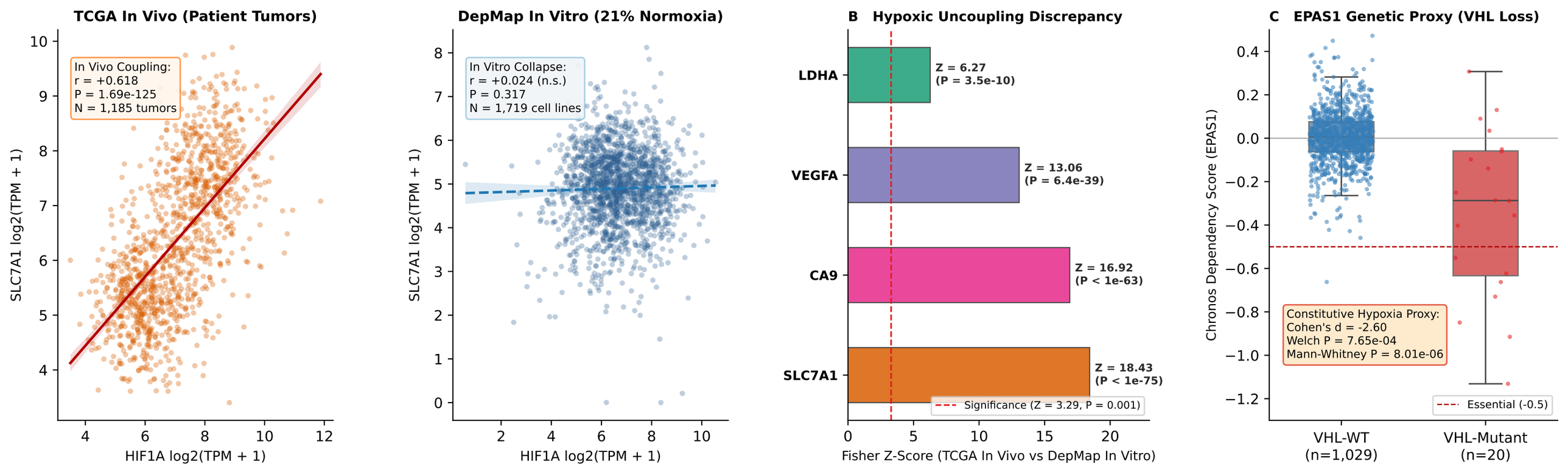

TCGA In Vivo (Patient Tumors)
In Vivo Coupling:
r = +0.618
P = 1.69e-125
N = 1,185 tumors
SLC7A1 log2(TPM + 1)
HIF1A log2(TPM + 1)
DepMap In Vitro (21% Normoxia)
In Vitro Collapse:
r = +0.024 (n.s.)
P = 0.317
N = 1,719 cell lines
SLC7A1 log2(TPM + 1)
HIF1A log2(TPM + 1)
B Hypoxic Uncoupling Discrepancy
LDHA
Z = 6.27
(P = 3.5e-10)
VEGFA
Z = 13.06
(P = 6.4e-39)
CA9
Z = 16.92
(P < 1e-63)
SLC7A1
Z = 18.43
(P < 1e-75)
Significance (Z = 3.29, P = 0.001)
Fisher Z-Score (TCGA In Vivo vs DepMap In Vitro)
C EPAS1 Genetic Proxy (VHL Loss)
Chronos Dependency Score (EPAS1)
Constitutive Hypoxia Proxy:
Cohen's d = -2.60
Welch P = 7.65e-04
Mann-Whitney P = 8.01e-06
Essential (-0.5)
VHL-WT
(n=1,029)
VHL-Mutant
(n=20)